



Baire's Theorem

Gabe Eze and Ben Wafford

Olin, Department of Mathematics
Centre College
Danville, KY

5/18/25

OUTLINE

1. **Definitions**
2. **Baire's Theorem**
3. **Metric Space Generalization and Implications**

Definitions

Definition 3.5.1. A set $A \subseteq \mathbb{R}$ is called an F_σ set if it can be written as the countable union of closed sets. A set $B \subseteq \mathbb{R}$ is called a G_δ set if it can be written as the countable intersection of open sets.

Definition 3.5.3. A set E is *nowhere-dense* if $\overline{E}$ contains no nonempty open intervals.

Baire's Theorem

Theorem 3.5.4. The set of real numbers $\mathbb{R}$ cannot be written as the countable union of nowhere-dense sets.

Proof.

For contradiction, assume that $E_1, E_2, E_3, \dots$ are each nowhere-dense and satisfy $\mathbb{R} = \bigcup_{n=1}^{\infty} E_n$. Then, ... □

More Definitions

Definition 8.2.4. A metric space (X, d) is *complete* if every Cauchy sequence in X converges to an element in X .

The definitions of F_σ , G_δ , and nowhere-dense sets extend to any metric space (X, d) .

- A set $E \subseteq X$ is an F_σ set if

$$E = \bigcup_{n=1}^{\infty} F_n,$$

where each F_n is closed in X .

- A set $G \subseteq X$ is a G_δ set if

$$G = \bigcap_{n=1}^{\infty} U_n,$$

where each U_n is open in X .

More Definitions

- **Definition 8.2.9.** A set $A \subseteq X$ is *dense* in the metric space (X, d) if $\overline{A} = X$. A subset E of a metric space (X, d) is *nowhere-dense* in X if $\overline{E}^\circ$ is empty.

Baire Category Theorem *A complete metric space is not the union of a countable collection of nowhere-dense sets.*

Implications

Set	Cardinality	Baire Category
$\mathbb{Q}$	Countable	First
Cantor	Uncountable	First
$\mathcal{C}[0, 1]$	Same as $\mathbb{R}$	Second
Differentiable at least at 1 point	Same as $\mathbb{R}$	First
Nowhere differentiable	Same as $\mathbb{R}$	Second

Table: Measuring "Size" in Real Analysis.